

Machine Learning Evaluation and Performance Metrics

Evaluating machine learning models requires appropriate metrics aligned with problem objectives. For regression, Mean Squared Error (MSE) penalizes large errors, Mean Absolute Error (MAE) provides interpretable average error magnitude, and R-squared indicates proportion of variance explained. Classification metrics include accuracy (correct predictions divided by total), precision (correct positive predictions divided by all positive predictions), recall (correct positive predictions divided by actual positives), and F1-score balancing precision and recall. Area Under the Receiver Operating Characteristic Curve (AUC-ROC) evaluates classification performance across thresholds. Confusion matrices reveal false positive and false negative rates. Class imbalance, where one class dominates, requires metric selection beyond accuracy. Cross-validation splits data into multiple folds, training and evaluating on different combinations, reducing variance in performance estimates. Learning curves plot training and validation performance versus data size, revealing underfitting (high bias) or overfitting (high variance). Hyperparameter tuning uses grid search or random search to optimize model performance. Regularization techniques like L1 and L2 penalty terms prevent overfitting by constraining model complexity. Early stopping halts training when validation performance stops improving. Ensemble methods combine multiple models for improved performance. Proper evaluation prevents deploying poorly generalizing models, ensuring real-world success.